

Question

Under certain conditions, a stoichiometric oxide A_xO_y crystal can be obtained. This crystal belongs to the cubic crystal system, and both **A** and O have only one type of coordination environment. Its structure can be described in two ways:

1. In this crystal, **A** forms a certain type of closest packing, and O orderly fills a certain proportion of a specific type of regular polyhedral voids.
2. This crystal can be regarded as a layered stacking, which is obtained by stacking **A** layers and O layers in a certain order.

The density of this crystal is $\rho = 3.22\text{g/cm}^3$, and the shortest distance between two O atoms in the crystal is $d = 229.4\text{pm}$.

Deduce the composition of the crystal and select the correct description from the following.

- A.** The ratio of the number of **A** layers to O layers in the crystal is 2 : 1.
- B.** Under normal temperature and pressure, A_xO_y is sparingly soluble in water.
- C.** The crystal has a face-centered cubic lattice.
- D.** The coordination number of **A** in this crystal is 4.
- E.** The shortest interlayer spacing between two O layers in the crystal, denoted as h (pm), satisfies $75 < h < 80$.
- F.** In this crystal, the O layer is a non-close-packed layer, where the occupancy ratio r of O satisfies $25\% < r < 75\%$.

Answer

Correct Answer: **E**

Detailed Explanation

Since the crystal has a cubic lattice, A must be cubic close-packed with 4 A atoms per unit cell. We can initially set $x = 4$.

CHECKPOINT

1 PTS

A is cubic close-packed

Let the unit cell parameter be a (pm), and the formula weight of the entire unit cell be M .

If O occupies the octahedral voids, then $d = a$ or $a/\sqrt{2}$.

If O occupies the tetrahedral voids, then $d = a/2$, $a/\sqrt{2}$, or $\sqrt{3}a/2$.

CHECKPOINT

4 PTS

d possible values are $a, a/\sqrt{2}, a/2, \sqrt{3}a/2$

The following scenarios are discussed case by case.

If $d = a$, then $y = 1$, $M = \rho a^3 N_A = 23.4(g/mol)$, which is unreasonable.

CHECKPOINT

1 PTS

$d = a$ is unreasonable

If $d = a/\sqrt{2}$, then $y = 4$, $M = 66.2(g/mol)$, which is unreasonable.

CHECKPOINT

1 PTS

$d = a/\sqrt{2}$ is unreasonable

If $d = a/2$, then $y = 8$, $M = 187.3(g/mol)$, which is unreasonable.

CHECKPOINT

1 PTS

$d = a/2$ is unreasonable

If $d = \sqrt{3}a/2$, then $y = 2$, $M = 36.0(g/mol)$, indicating A is H, so the crystal is H_4O_2 , i.e., H_2O .

CHECKPOINT

2 PTS

$d = \sqrt{3}a/2$ has a reasonable solution H_2O

In reality, this crystal adopts the Cu_2O packing, with the ratio of A layers (H layers) to O layers being 1 : 2, so option A is incorrect.

CHECKPOINT

3 PTS

The ratio of A layers to O layers is 1 : 2

H_2O is not poorly soluble in water, so option B is incorrect.

CHECKPOINT

1 PTS

H_2O is not poorly soluble in water

This crystal has a simple cubic lattice, so option C is incorrect.

CHECKPOINT

1 PTS

This crystal has a simple cubic lattice

In this crystal, the coordination number of A (H) is 2, so option D is incorrect.

CHECKPOINT

1 PTS

The coordination number of A is 2

In this crystal, the interlayer spacing of O layers is $h = \sqrt{3}a/6 = 76.5pm$, so option E is correct.

CHECKPOINT

3 PTS

The interlayer spacing of O layers is $h = 76.5pm$

In this crystal, the occupancy rate of O layers is 25%, so option F is incorrect.

CHECKPOINT

1 PTS

The occupancy rate of O layers is 25%